

Basic Analysis by Jiri Lebl

Exercise 3.1.10

Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $\forall \{x_n\}_{n=1}^{\infty}$ in $\mathbb{R}$, the sequence $\{f(x_n)\}_{n=1}^{\infty}$ converges. Prove that f is constant, that is, $f(x) = f(y)$ for all $x, y \in \mathbb{R}$.

Proof.

First let us construct a sequence $\{x_n\}_{n=1}^{\infty}$ that is equal to some value $a \in \mathbb{R}$ when n is odd and equal to $b \in \mathbb{R}$ when n is even.¹

We know that $\{f(x_n)\}_{n=1}^{\infty}$ converges, so given any $\epsilon > 0$, $\exists N$ such that $|f(x_n) - f(x_m)| < \epsilon$, $\forall n, m \geq N$. Since $f(x_n)$ can only be $f(a)$ or $f(b)$, there must be some $n, m \geq N$ where $f(x_n) = f(a)$ and $f(x_m) = f(b)$ (or vice versa). So $|f(a) - f(b)| < \epsilon$ for any $\epsilon > 0$, which means $f(a) = f(b)$.

And since the sequence $\{x_n\}_{n=1}^{\infty}$ can be constructed for any $a, b \in \mathbb{R}$ then the function f is constant.

□

¹A formula for such a function can be $x_n = \max(0, (-1)^{n-1})a + \max(0, (-1)^n)b$ or $x_n = \frac{1}{2}(a + b + (-1)^n(b - a))$